Molecular Dynamics Simulation Report

Project: **protein_ligand_auto**

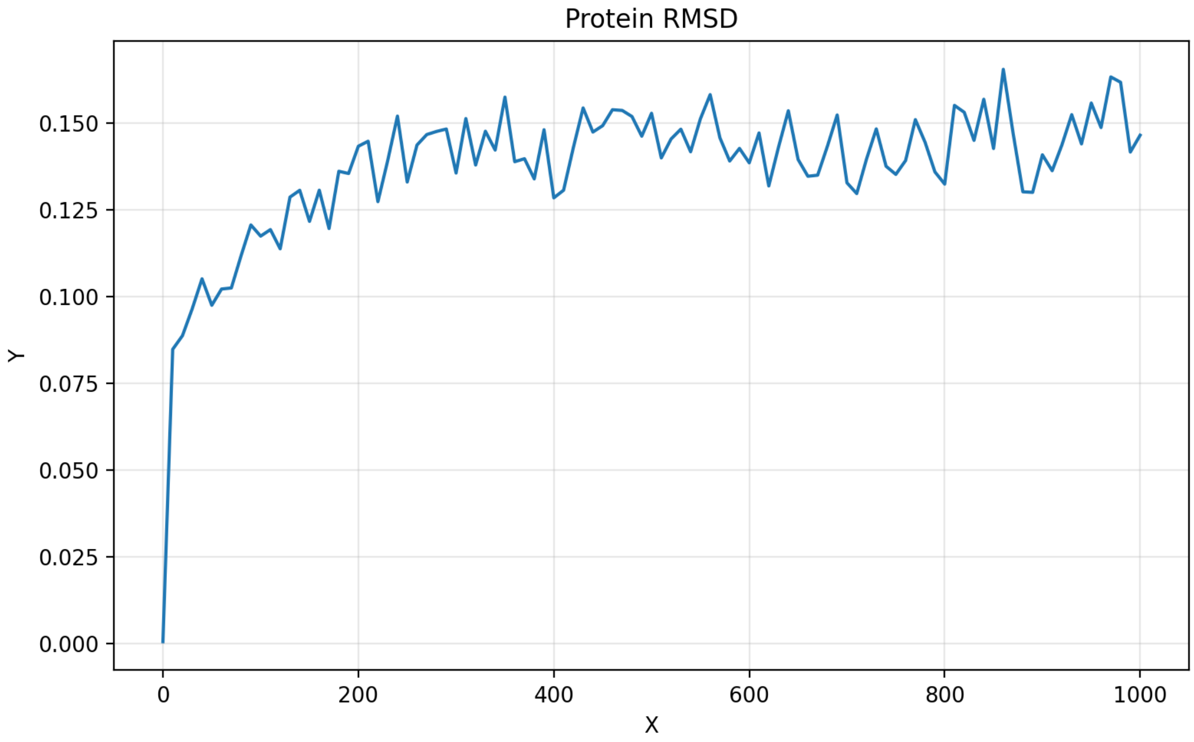
Generated: **2026-08-20 19:45:18**

1. Simulation Parameters

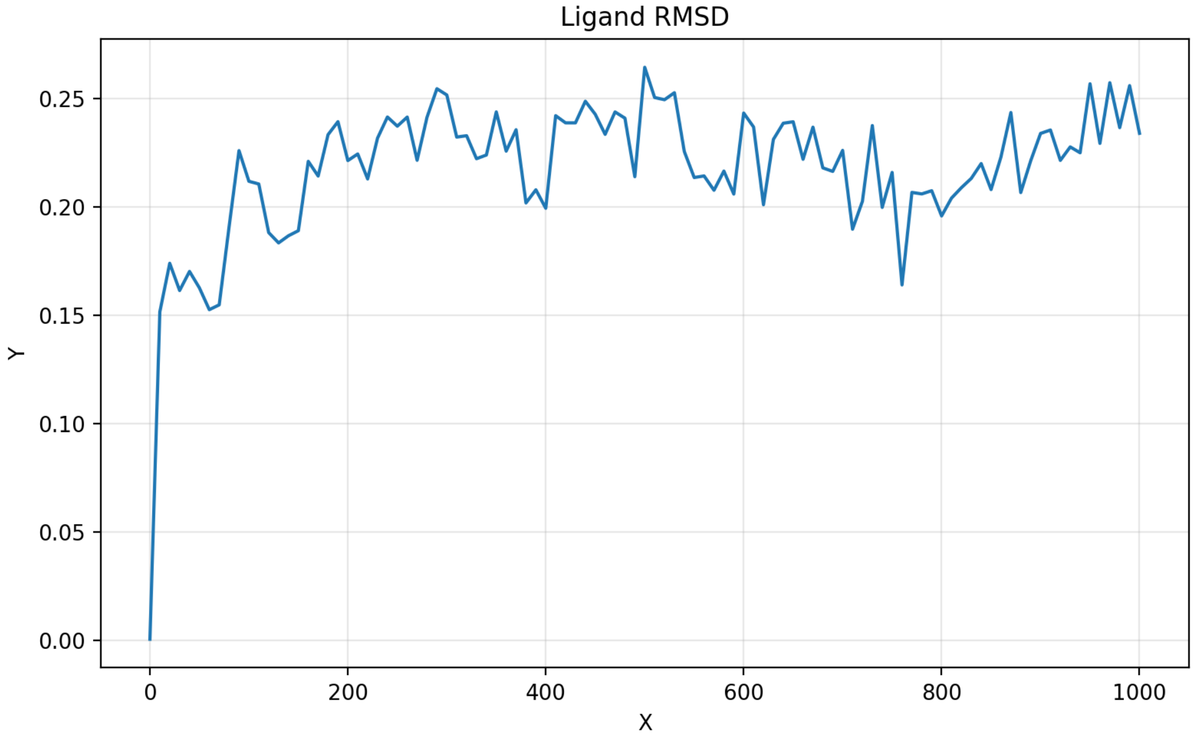
Parameter	Value
Force Field	charmm36-feb2026_cgenff-5.0
Water Model	tip3p
Ligand Prep Tool	acpype
Ligand Residue	LIG
Production Time (ns)	1
Temperature (K)	300
Pressure (bar)	1.0
Timestep (ps)	0.002
Electrostatics	PME
VDW Method	Force-switch
Cutoff (nm)	1.2
TCoupl	V-rescale
PCoupl	Parrinello-Rahman
Constraints	H-bonds (LINCS)
GPU Acceleration	auto
CPU Threads	auto

2. Structural Analysis

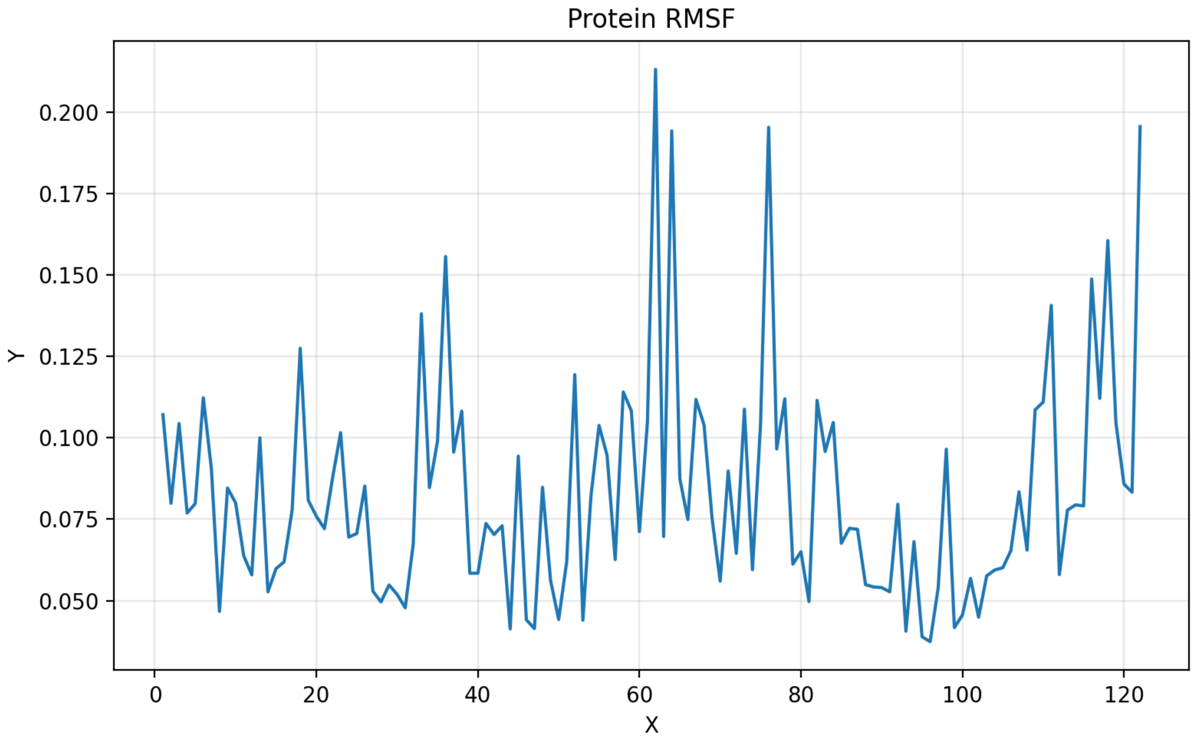
2.1 RMSD - Protein



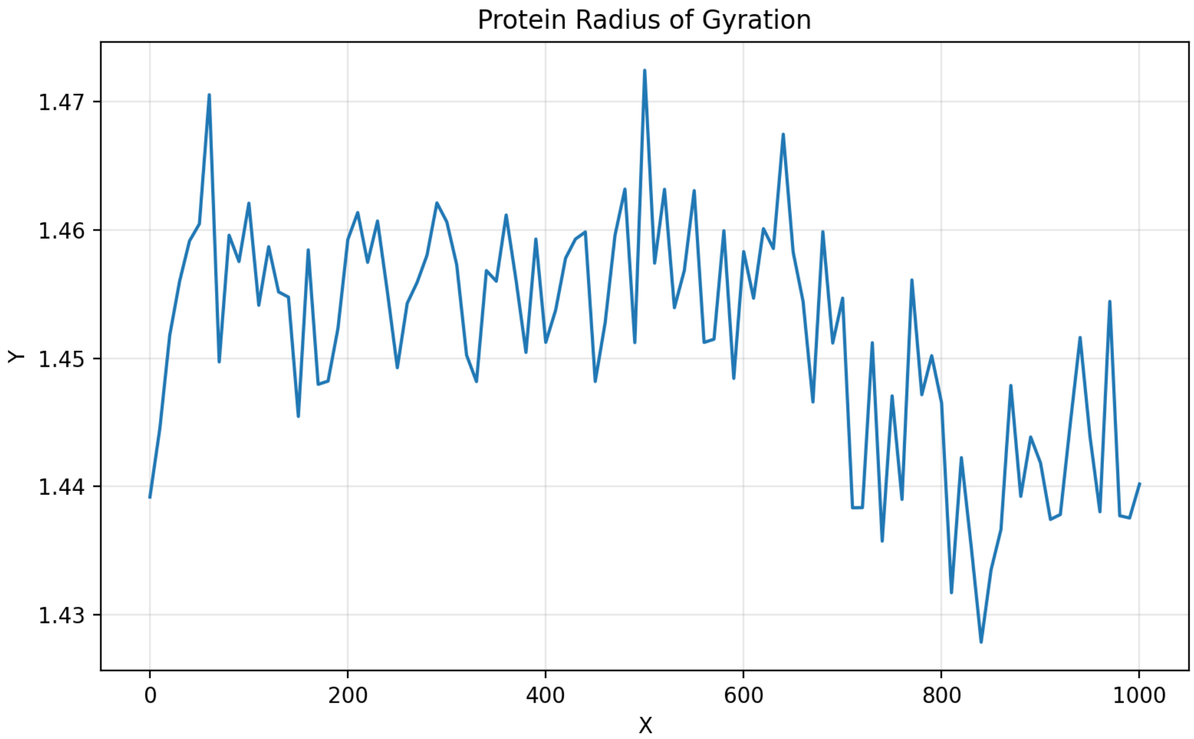
2.2 RMSD - Ligand



2.4 RMSF - Protein

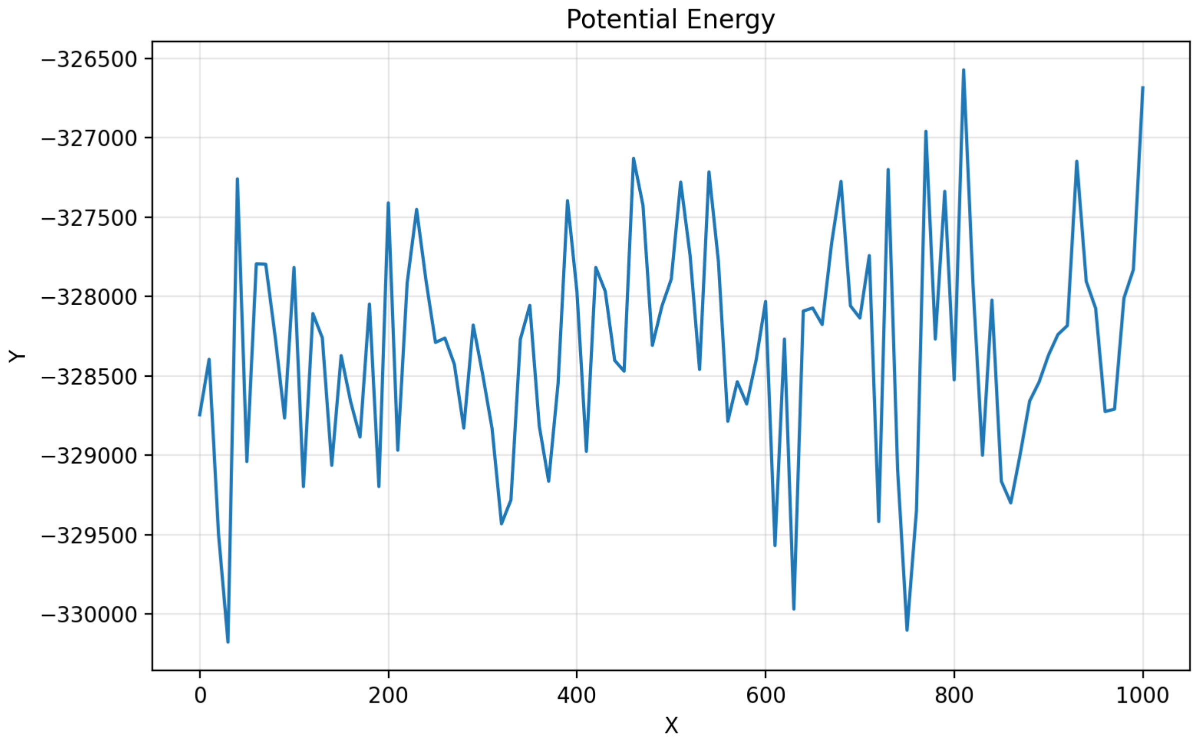


2.5 Radius of Gyration

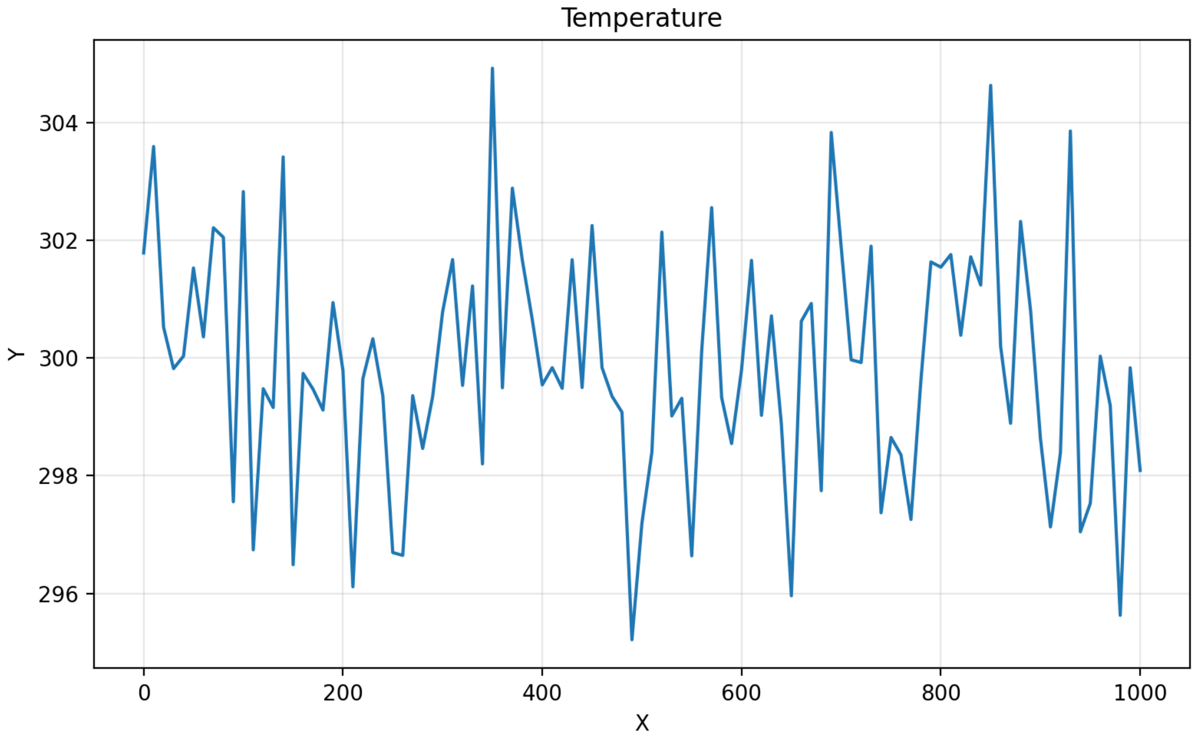


3. Energetics

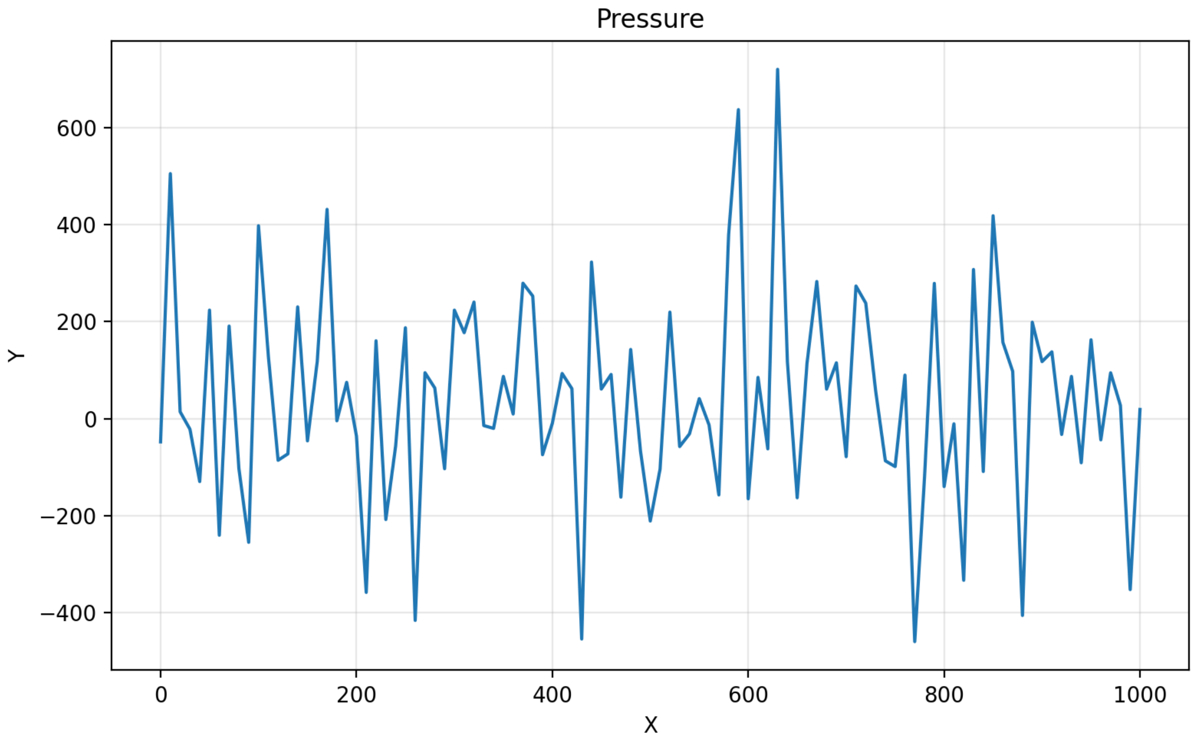
3.1 Potential Energy



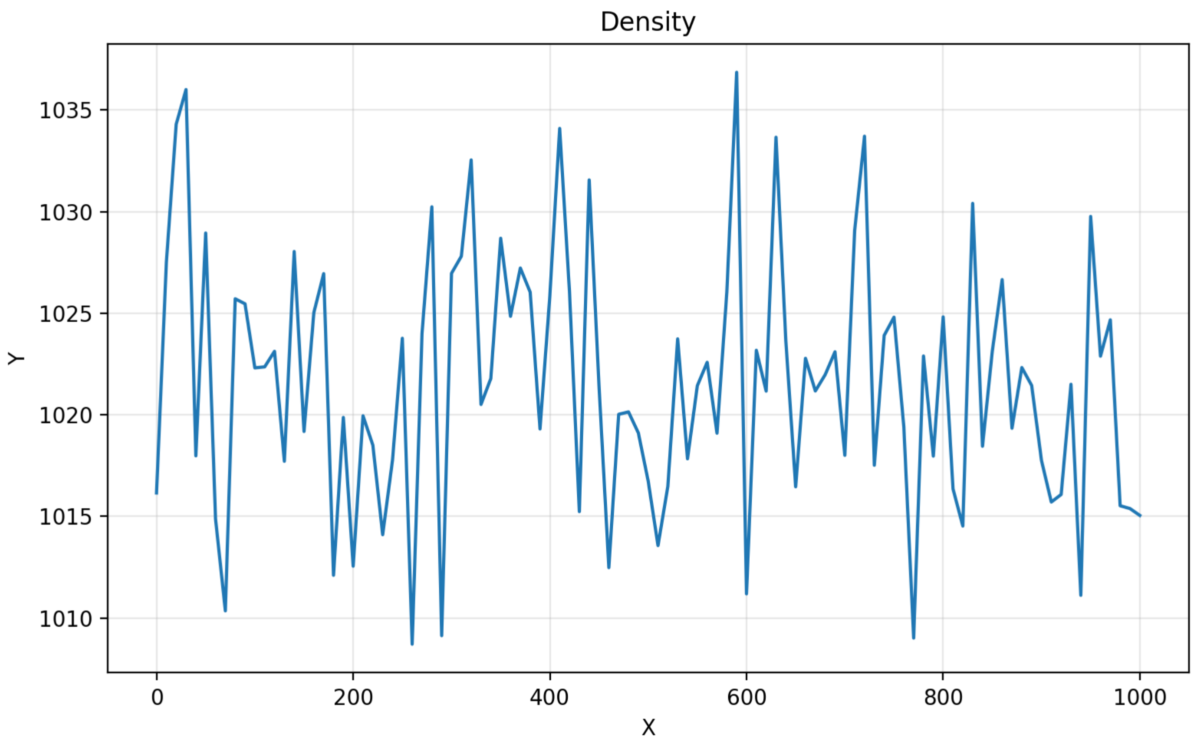
3.2 Temperature



3.3 Pressure


Pressure
Y
X
600
400
200
0
−200
−400
0
200
400
600
800
1000

3.4 Density


Density
Y
X
1035
1030
1025
1020
1015
1010
0
200
400
600
800
1000

4. MM-PBSA Binding Free Energy

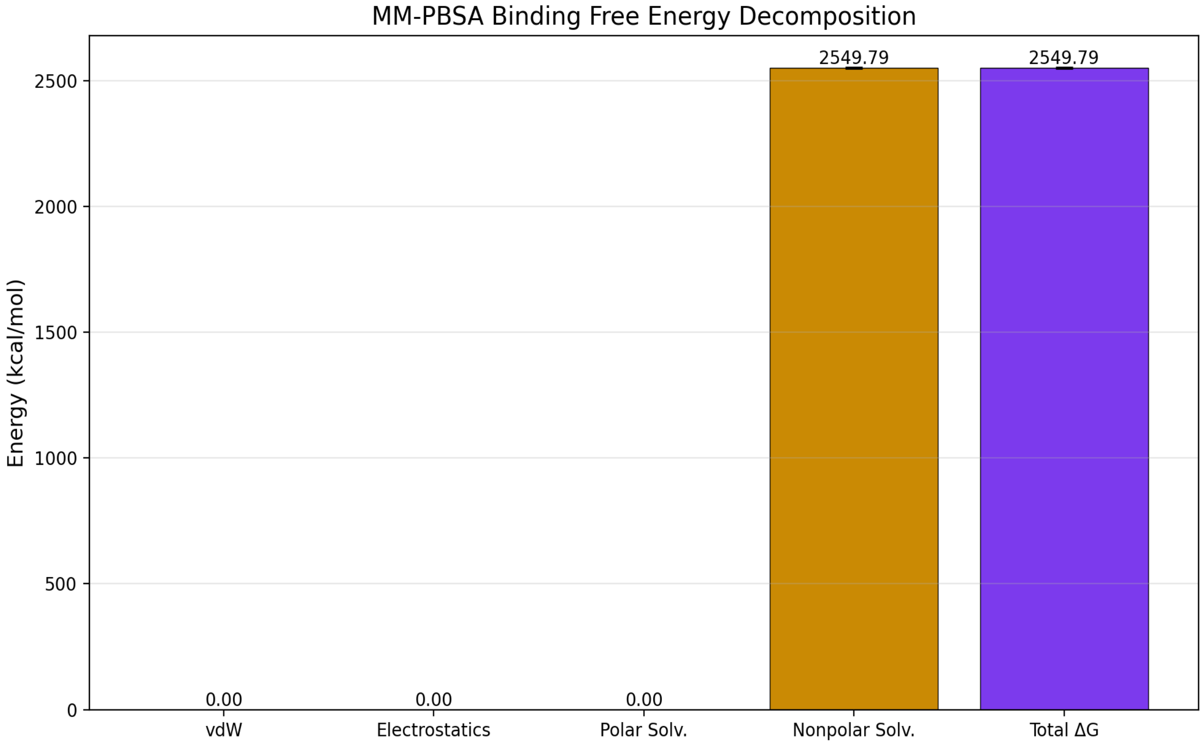
4.1 MM-PBSA Parameters

Parameter	Value
Energy Method	GROMACS (force-field parameters from TPR)
PB Method	APBS Poisson-Boltzmann
Frames Analyzed	11
Frame Step	10
Temperature (K)	300.0
pH	7.0
Ionic Strength (M)	0.15
Surface Tension	0.0072
Receptor Selection	protein
Ligand Selection	index 1823 1824 1825 1826 1827 1828 1829 1830 1831 1832 1833 1834 1835 1836 1837 1838 1839 1840 1841 1842 1843 1844 1845 1846 1847 1848 1849 1850 1851 1852 1853 1854 1855 1856 1857 1858 1859 1860 1861 1862 1863 1864 1865 1866 1867 1868 1869 1870 1871 1872 1873 1874 1875 1876 1877 1878 1879 1880 1881 1882 1883 1884 1885 1886 1887 1888 1889 1890 1891 1892 1893 1894 1895 1896 1897 1898 1899

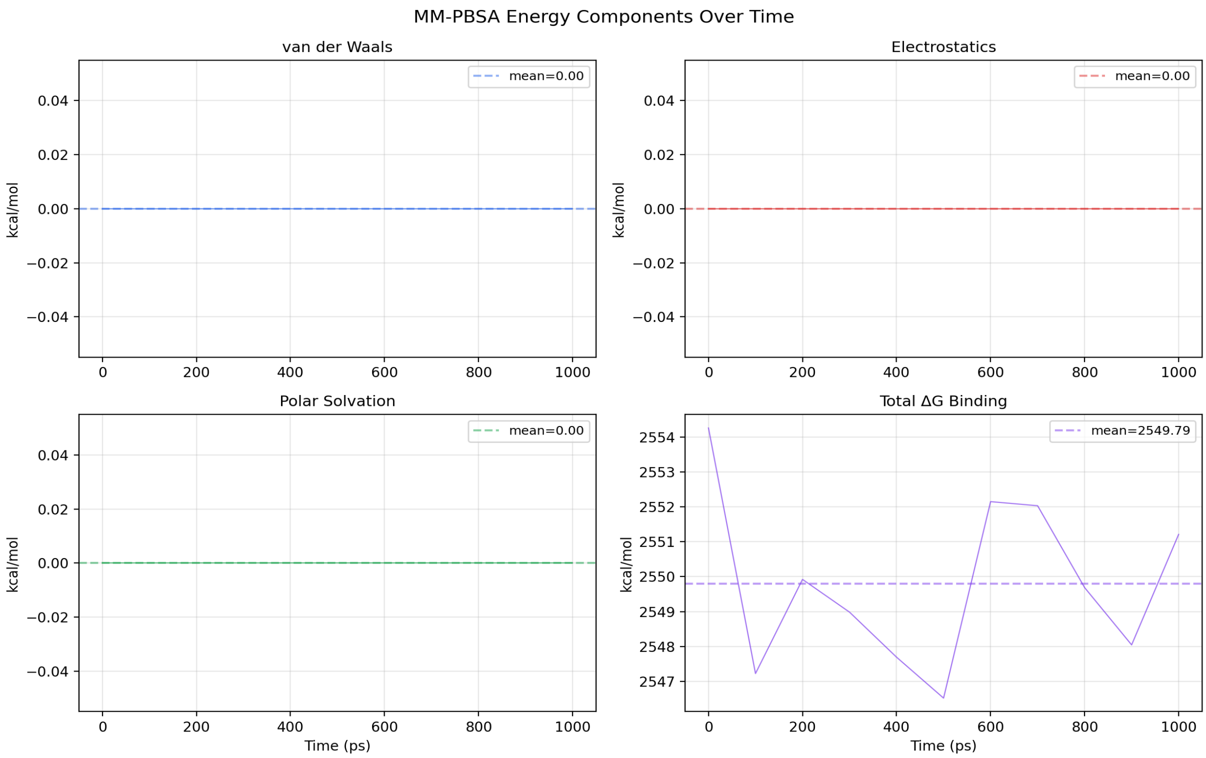
4.2 Binding Free Energy Decomposition

Component	Mean (kcal/mol)	Std Dev	SEM
van der Waals	0.0000	0.0000	0.0000
Electrostatics	0.0000	0.0000	0.0000
Polar Solvation	0.0000	0.0000	0.0000
Nonpolar Solvation	2549.7932	2.2959	0.6922
Total ΔG Binding	2549.7932	2.2959	0.6922

4.3 Energy Component Bar Chart

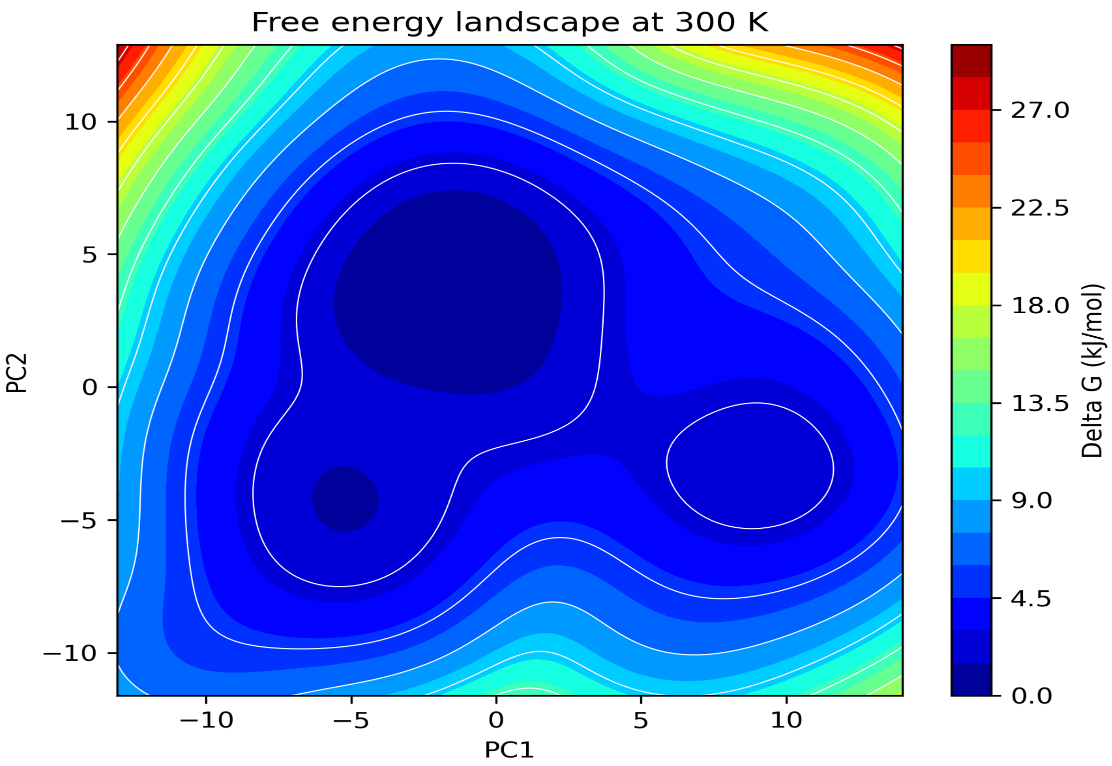


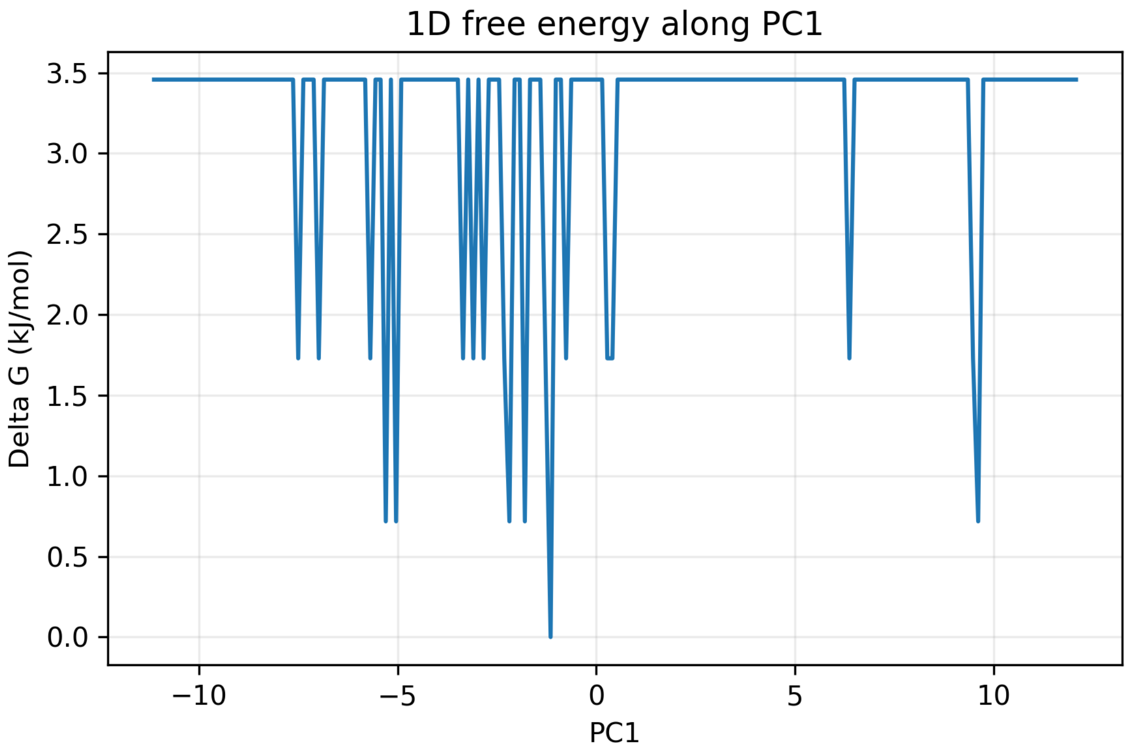
4.4 Energy Components Over Time

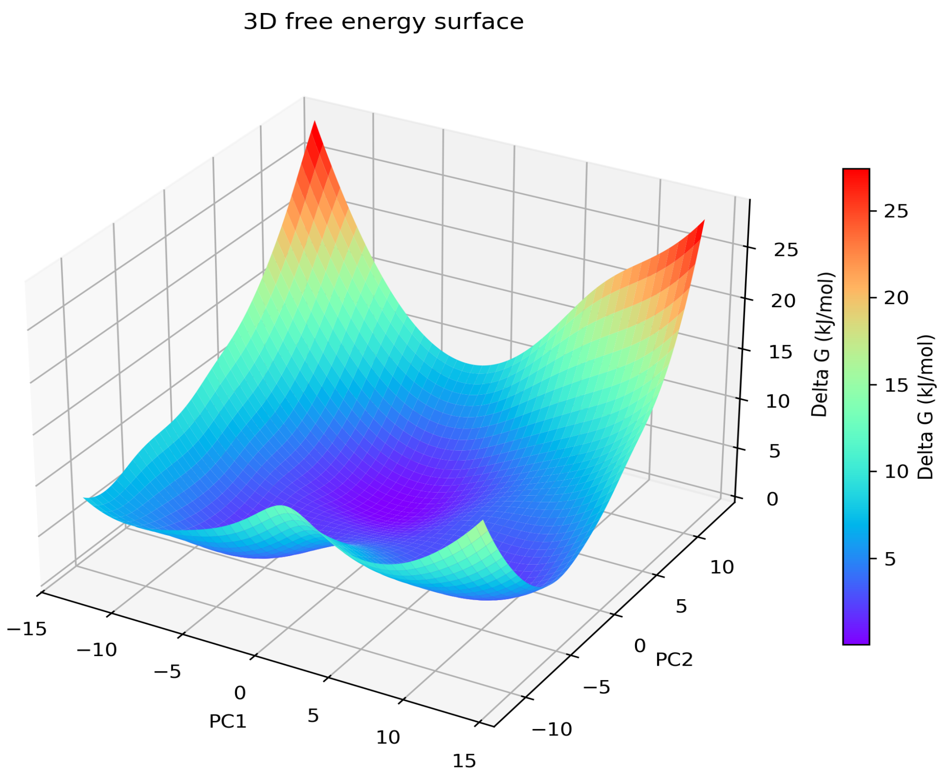


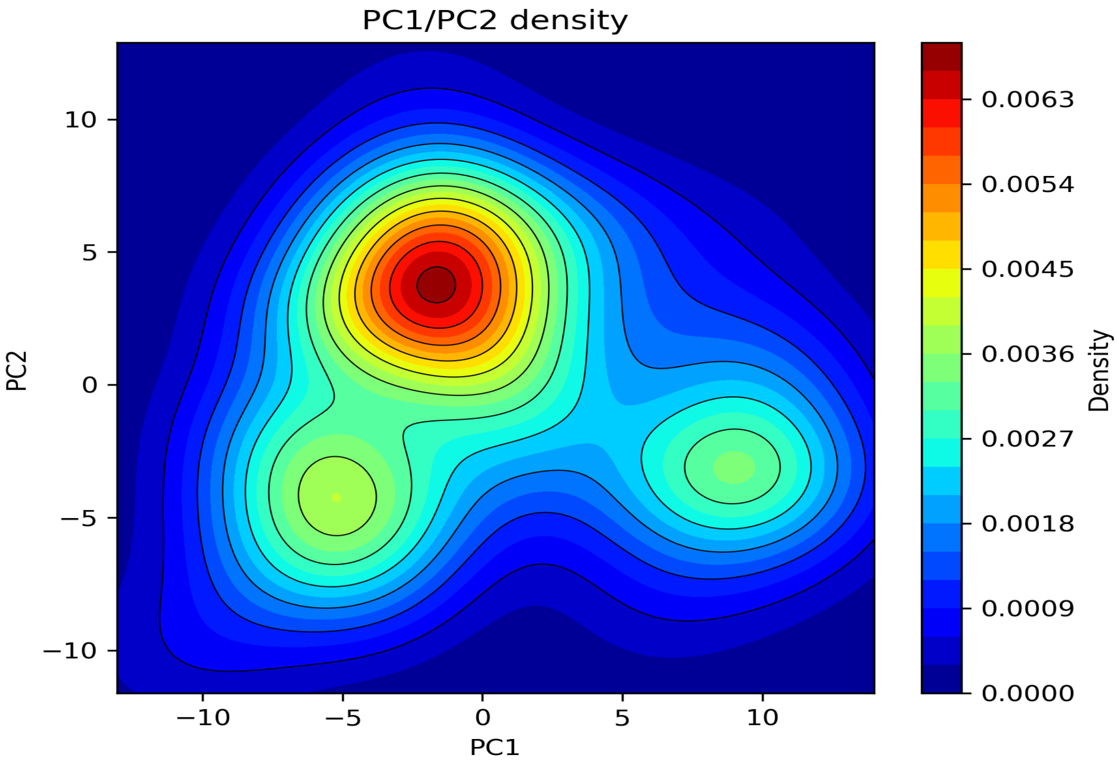
5. Free Energy Landscape

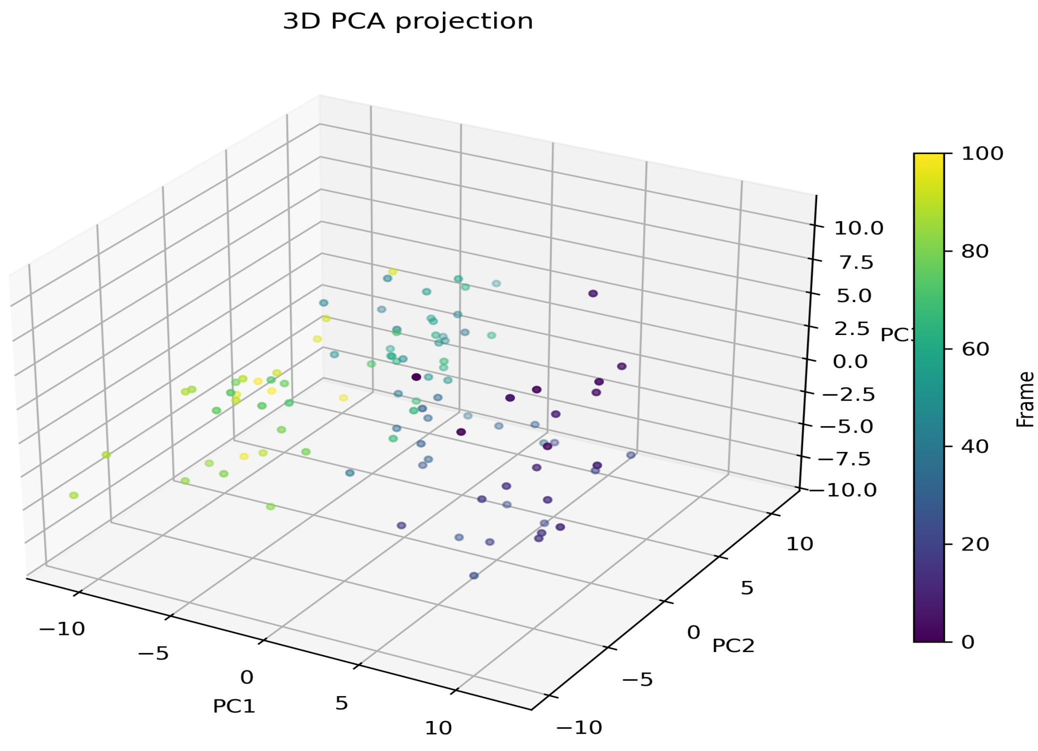
Parameter	Value
Selection	backbone
Temperature (K)	N/A
PCA Components	N/A
Grid Bins	N/A

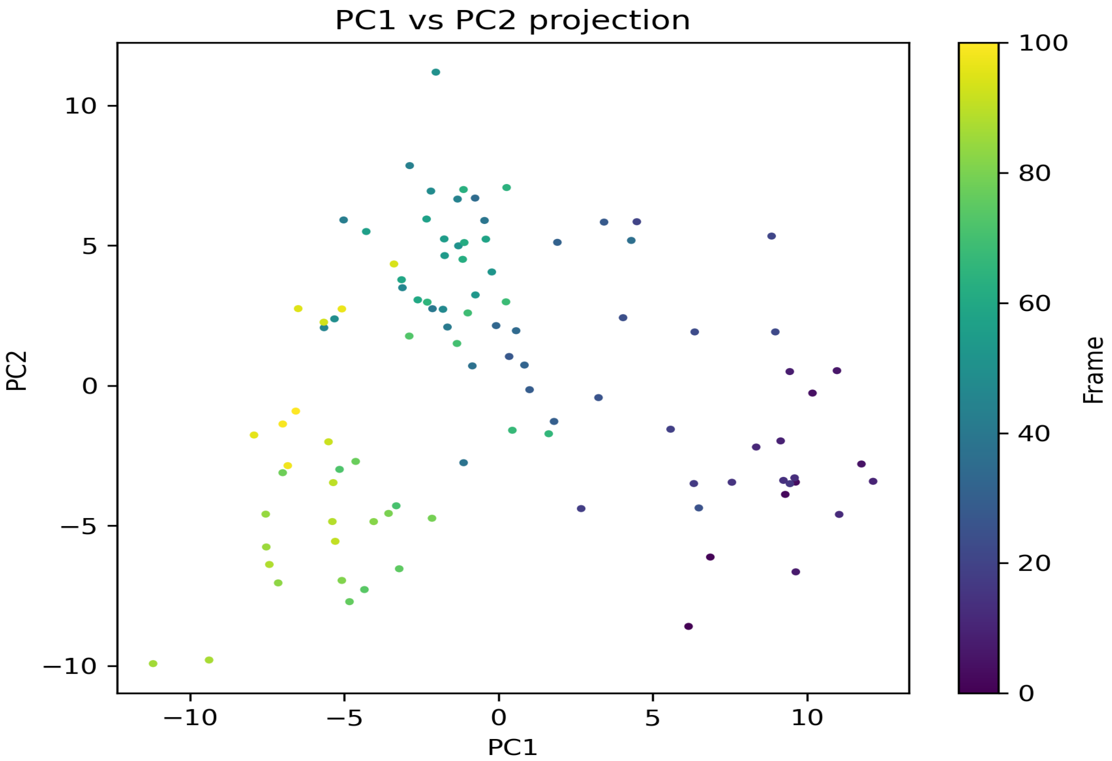


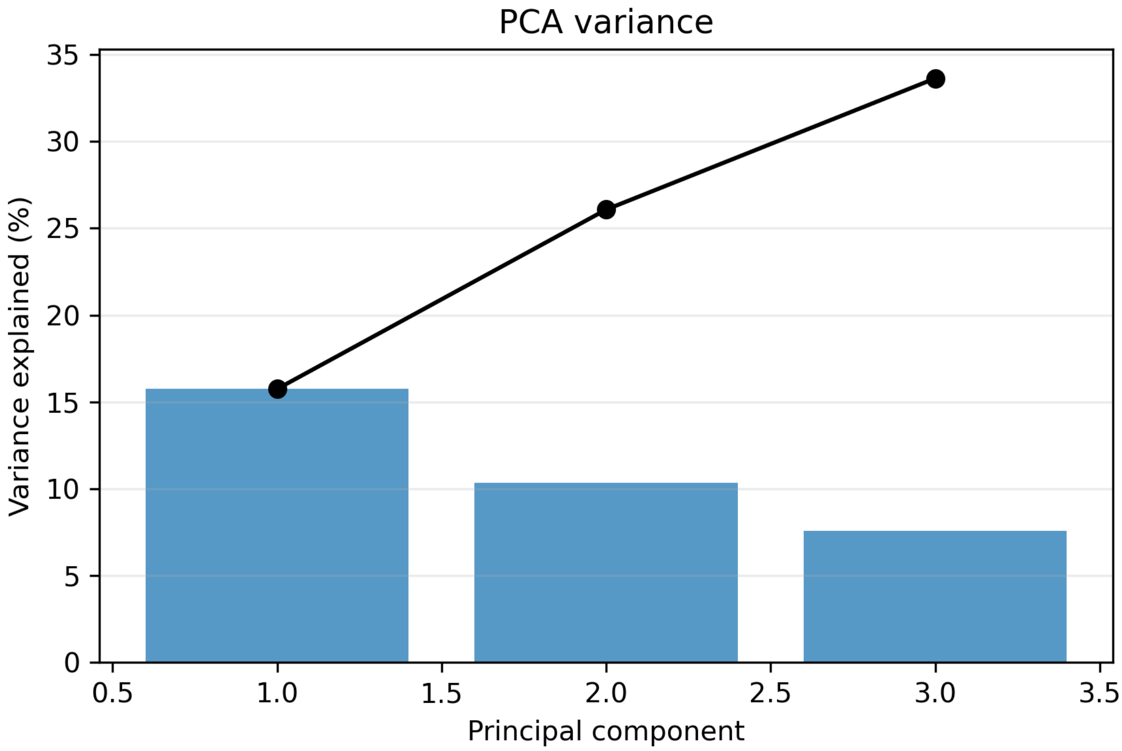

1D free energy along PC1
Delta G (kJ/mol)
PC1
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−10
−5
0
5
10


3D free energy surface
PC1
PC2
Delta G (kJ/mol)
Delta G (kJ/mol)
−15
−10
−5
0
5
10
15
25
20
15
10
5
0


PC1/PC2 density
PC2
PC1
Density
0.0063
0.0054
0.0045
0.0036
0.0027
0.0018
0.0009
0.0000


3D PCA projection
PC1
PC2
Frame


PC1 vs PC2 projection
PC1
PC2
Frame
10
5
0
−5
−10
0
20
40
60
80
100


PCA variance
Variance explained (%)
Principal component
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0
5
10
15
20
25
30
35

6. Summary

This report summarizes the molecular dynamics simulation results generated by the automated MD pipeline. All analyses were performed using GROMACS tools and custom Python scripts. The MM-PBSA binding free energy was calculated using APBS for Poisson-Boltzmann electrostatics and MDAnalysis for trajectory processing.

Estimated Binding Free Energy (ΔG): 2549.79 ± 2.30 kcal/mol

For questions about this report or the analysis methodology, please refer to the pipeline documentation or contact the project administrator.